Implicit Differentiation and Inverse Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

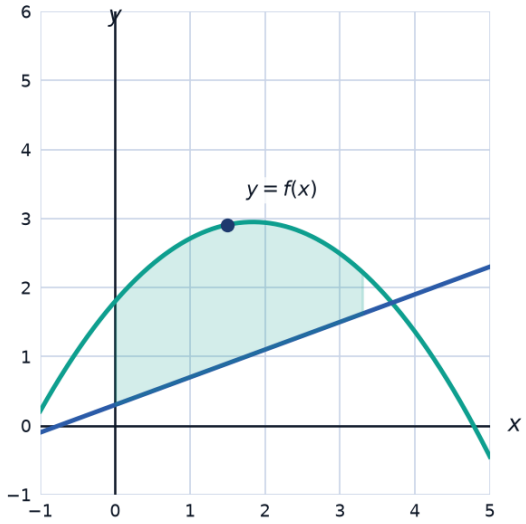
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · MediumDuring Section II Part A (30 minutes, Calculator Active), a student completes Questions 1 and 2 with 5 minutes remaining. When time is called and calculators are stored for Part B (60 minutes, No Calculator), may the student return to Questions 1 and 2 to add work, and what restrictions apply?

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Multiple Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Vectors & Vector-Valued Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Unit 9 · Parametric Equations & Vector-Valued Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

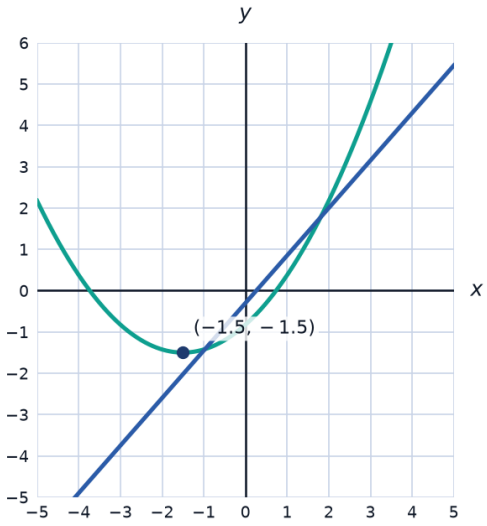
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[AP Calculus BC Exam Format & Units Overview · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Functions (HL Extensions) · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 3 · Advanced For the position vector $\mathbf{r}(t) = \langle t, t^2, \frac{2}{3}t^3 \rangle$, find the tangential component of acceleration a_T and the normal component a_N at $t = 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

(IB AA/AI HL) Given the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's numerical method with step size $h = 0.1$ to approximate $y(0.2)$. Compare your numerical result with the exact analytical solution.

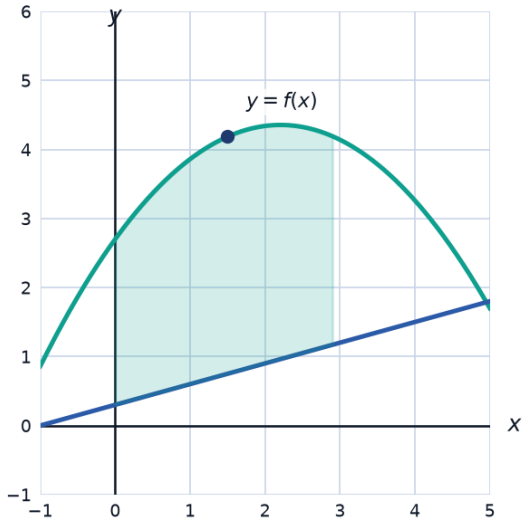
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 3 · Advanced / AP FRQOn an AP Calculus BC exam question asking whether a series $\sum_{n=1}^{\infty} a_n$ converges, a student writes: "Since $\lim_{n\to\infty} a_n = 0$, the series converges." Evaluate the mathematical validity of this statement, identify the logical fallacy, and provide a standard counterexample that disproves it.

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $\mathbf{a} = (2;\ -1;\ 3)$ and $\mathbf{b} = (1;\ 4;\ -2)$. Find the cross product $\mathbf{a} \times \mathbf{b}$, and determine the exact area of the triangle with adjacent sides $\mathbf{a}$ and $\mathbf{b}$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the exact range of $f(x) = \frac{x}{x^2 + 1}$ for $x \in \mathbb{R}$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 3 · Advanced / AP FRQLet $y = f(x)$ be the particular solution to $\frac{dy}{dx} = xy^2$ with $f(1) = -1$. Use Euler's method with step size $\Delta x = 0.5$ starting at $x = 1$ to approximate $f(2)$, and determine whether the approximation is an overestimate or underestimate by computing $\frac{d^2y}{dx^2}$ at $(1,\ -1)$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · Medium Find the area of the triangle with vertices $A(1, 1, 0)$, $B(3, 0, 2)$, and $C(0, 2, 4)$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · Advanced Evaluate the double integral $\iint_D e^{x^2 + y^2}\, dA$ where D is the semi-annular region $\{(x, y) : 1 \le x^2 + y^2 \le 4, y \ge 0\}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

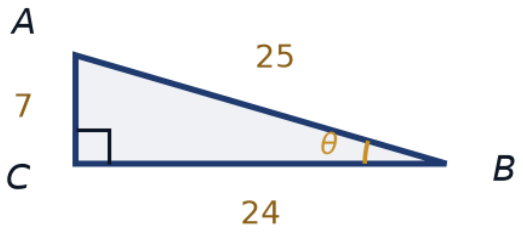
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Level 2 · Medium Evaluate the iterated integral by reversing the order of integration: $\int_0^{\pi} \int_y^{\pi} \frac{\sin x}{x}\, dx\, dy$

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ of $y = x^{\sin x}$ for $x > 0$ using logarithmic differentiation.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Level 2 · MediumLet $\frac{dy}{dx} = x + 2y$ with $y(0) = 1$. Use Euler's method with step size $\Delta x = 0.2$ to approximate $y(0.4)$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Level 1 · Foundations Evaluate $\iint_R (2x + y)\,dA$ where R is the rectangular region $[0, 3] \times [1, 2]$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Level 2 · MediumA particle moves with position vector $\vec{r}(t) = \langle t^2 - 4t, \frac{1}{3}t^3 - 4t \rangle$. At what time $t > 0$ is the particle at rest?

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

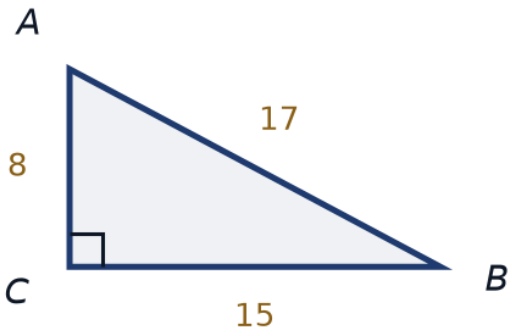
Level 3 · Advanced / AP FRQA particle moves along a planar path with velocity $\vec{v}(t) = \langle \sin(t^2), \cos(t^3) \rangle$ for $0 \le t \le 3$. Write the definite integral representing the total distance traveled by the particle, and calculate the numerical value using three decimal places.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the slope of the tangent line to $x^2y + y^2 = 5$ at the point $(2, 1)$.



a. $-\frac{2}{3}$

b. $-\frac{4}{5}$

c. $-\frac{4}{6}$

d. -1

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For $x^2 + y^2 = 16$, what is $\frac{d^2y}{dx^2}$ expressed solely in terms of y?

a. $-\frac{16}{y^3}$

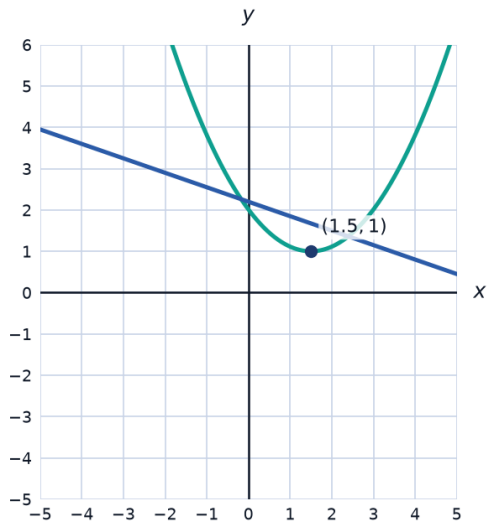
b. $\frac{16}{y^3}$

c. $-\frac{1}{y^2}$

d. $-\frac{4}{y^3}$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

At the point $(0, 2)$ on the ellipse $4x^2 + y^2 = 4$, what is $\frac{d^2y}{dx^2}$?



a. -2

b. 2

c. -4

d. 0

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the sign of the concavity at $(3, 4)$ on the circle $x^2 + y^2 = 25$?

a. Negative (concave down)

b. Positive (concave up)

c. Zero (inflection point)

d. Undefined

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If g is the inverse of f, which formula gives $g'(a)$ where $f(b) = a$?

a. $g'(a) = \frac{1}{f'(b)}$

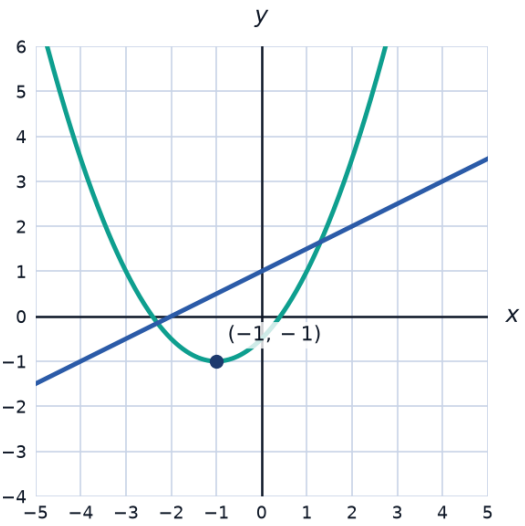
b. $g'(a) = f'(b)$

c. $g'(a) = \frac{1}{f'(a)}$

d. $g'(a) = -\frac{1}{f'(b)}$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = x^3 + 2x - 1$. If g is the inverse of f, find $g'(2)$.



a. $\frac{1}{5}$

b. 5

c. $\frac{1}{14}$

d. 14

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Let $f(x) = x^5 + x + 1$. What is the derivative of f^{-1} at $x = 3$?

a. $\frac{1}{6}$

b. 6

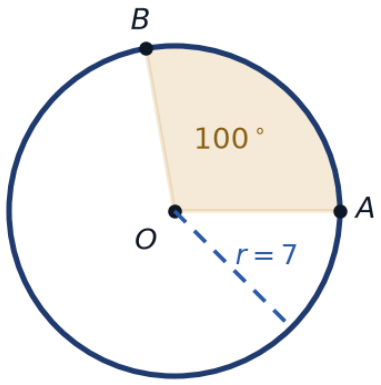
c. $\frac{1}{80}$

d. $\frac{1}{5}$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the slope of the tangent line to $y = \arctan(x)$ at $x = 1$?



a. $\frac{1}{2}$

b. 1

c. $\frac{\pi}{4}$

d. $\frac{1}{\sqrt{2}}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find $\frac{dy}{dx}$ by implicit differentiation if $2x^2 + 3y^2 = 1$.

a. $\frac{2x}{3y}$

b. $-\frac{2x}{3y}$

c. $-\frac{3y}{2x}$

d. $-\frac{2x}{3}$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Implicit differentiation of $x^2 + y^2 = 25$ gives $\frac{dy}{dx} =$

a. $\frac{x}{y}$

b. $-\frac{x}{y}$

c. $\frac{y}{x}$

d. $-\frac{y}{x}$

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Identify the general solution to the first-order differential equation $\frac{dy}{dx} = 3y$.

a. $y = C\ln(3x)$

b. $y = \frac{3}{2}x^2 + C$

c. $y = Ce^{-3x}$

d. $y = Ce^{3x}$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the average value of $f(x) = 4x^2$ on $[3, 7]$.

a. $\frac{1264}{3}$

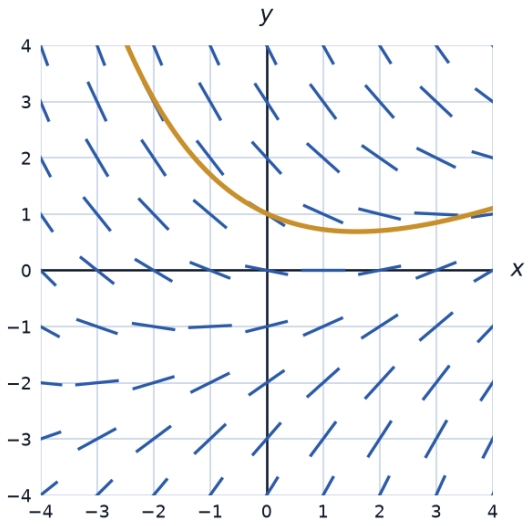
b. $\frac{310}{3}$

c. $\frac{322}{3}$

d. $\frac{316}{3}$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A population grows according to the logistic model $\frac{dP}{dt} = 0.08P\left(1 - \frac{P}{315}\right)$. Find the population P at the point of maximum growth.



a. 630

b. 78.75

c. 315

d. 157.5

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given $f(x) = \{1x + 3, x < 5; 3x - 6, x \geq 5$, find $\lim_{x \to 5} f(x)$, if it exists.

a. does not exist

b. 8

c. 18

d. 9

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Classify the discontinuity of $f(x) = \frac{(x-2)(x-4)}{x-2}$ at $x = 2$.

a. infinite (a vertical asymptote)

b. removable (a hole)

c. the function is continuous there

d. jump discontinuity

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $G(x) = \int_5^{x^3} \sqrt{t+2}\, dt$. By the Fundamental Theorem of Calculus, what is $G'(x)$?

a. $3x^2\sqrt{x+2}$

b. $\sqrt{x^3+2}$

c. $3x^2\sqrt{x^3+2}$

d. $x^3\sqrt{x^3+2}$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For the differential equation $\frac{dy}{dx} = 1x - 5y$, find the slope of the solution curve at the point $(3, 2)$.

a. -9

b. -5

c. 7

d. -7

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

If $y = 6e^{1t}$ is a solution of $\frac{dy}{dt} = ky$, what is the value of k?

a. 2

b. 1

c. 3

d. -1

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Implicit differentiation: $\frac{d^2y}{dx^2}$ at $(1, 1)$ for $x^2 + y^2 = 2$:

a. -1

b. -2

c. 0

d. 1

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Implicit function theorem for optimization: constrained critical points satisfy $abla_{\mathbf{x}} L = 0$ where $L = f - \lambda g$. Name of λ:

a. Lagrange multiplier

b. eigenvalue

c. gradient

d. curvature

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

The magnitude of cross product $|\mathbf{a} \times \mathbf{b}|$ equals the area of the parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$. For $|a| = 3, |b| = 4, \theta = \pi/6$:

a. 24

b. 12

c. 3

d. 6

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Area Between Curves

Cross product $\mathbf{a} \times \mathbf{b}$ is orthogonal to:

a. $\mathbf{a}$ only

b. $\mathbf{b}$ only

c. neither

d. both $\mathbf{a}$ and $\mathbf{b}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Implicit differentiation of $xy = 5$: $\frac{dy}{dx} =$

a. $\frac{5}{x}$

b. $\frac{y}{x}$

c. $-\frac{y}{x}$

d. $\frac{x}{y}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Limits Graphically and Numerically

Integration of inverse functions (Laisant-Hermite): $\int_a^b f^{-1}(y)dy = bf^{-1}(b) - af^{-1}(a) - \int_{f^{-1}(a)}^{f^{-1}(b)} f(x)dx$. Example for $f = e^x$:

a. gives nothing useful

b. yields: $\ln b - \ln a$

c. gives $e^b - e^a$

d. yields: $b\ln b - a\ln a - (b - a)$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

The cross product of two parallel vectors is:

a. A unit vector

b. The zero vector

c. Their dot product

d. Undefined

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